THE MASTERMIND COACHING CLASSES

Revision Notes

Class 9 · Mathematics

Ganita Manjari — Part 1, Chapters 1 to 8 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Orienting Yourself: The Use of Coordinates	Cartesian plane, quadrants, reflections, distance formula, midpoint and section points, shapes on the plane
2	Introduction to Linear Polynomials	Linear patterns, polynomial terms, linear polynomials, graphs, slope and intercept, zeros, modelling
3	The World of Numbers	Number families, closure, rational numbers and decimals, irrational numbers, real number line
4	Exploring Algebraic Identities	Identities for squares and cubes, $(a + b + c)^2$, factorisation, quick arithmetic, Śhrīdharāchārya's method
5	Circles	Chords and the centre, equal chords, angle at the centre and circumference, cyclic quadrilaterals
6	Measuring Space: Perimeter and Area	Circumference and area of circles, arcs and sectors, area of triangles (Heron), parallelograms, trapezium and composite figures
7	The Mathematics of Maybe: Probability	Experimental and theoretical probability, sample space, equally likely outcomes, complementary events, law of large numbers
8	Predicting What Comes Next: Sequences and Progressions	Sequences, explicit and recursive rules, arithmetic and geometric progressions, sum of an AP, Virahānka and Sierpiński patterns

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CHAPTER 01 · MATHS (PART 1)

Orienting Yourself: The Use of Coordinates

Coordinates give every point in a plane an address, so geometry can be done with numbers. This chapter sets up the Cartesian plane, locates points and quadrants, reflects points in the axes, derives the distance and midpoint formulae and uses them to identify triangles, rectangles and collinear points.

Big Questions

- How is the position of a point fixed by an ordered pair?
- How do we find the distance between two points and the midpoint of a segment?
- How can coordinates prove that a figure is isosceles, a rectangle or a parallelogram, or that points are collinear?

1.1 The Cartesian Plane

DEFINITION

Two perpendicular number lines, the horizontal **x-axis** and the vertical **y-axis**, meet at the **origin O(0, 0)**. A point is named by an **ordered pair (x, y)**: **x = abscissa** (distance along the x-axis) and **y = ordinate** (distance along the y-axis). (3, 5) and (5, 3) are different points.

Region	Signs of (x, y)	Example
Quadrant I	(+, +)	(2, 5)
Quadrant II	(−, +)	(−2, 5)
Quadrant III	(−, −)	(−1, −1)
Quadrant IV	(+, −)	(6, −3)
On the x-axis	y = 0	(4, 0)
On the y-axis	x = 0	(0, −4)

REMEMBER

Points on the **axes** belong to **no quadrant**. The origin has coordinates (0, 0). A point 5 units below the origin on the y-axis is **(0, −5)**.

1.2 Reflections

Reflection of P(x, y) in	Image	Example (3, −2)
the x-axis	(x, −y)	(3, 2)
the y-axis	(−x, y)	(−3, −2)
the origin	(−x, −y)	(−3, 2)
the line y = x	(y, x)	(−2, 3)

1.3 Distance Between Two Points

FORMULA

Distance formula: for A(x_1, y_1) and B(x_2, y_2), **$AB = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$**. Distance of P(x, y) from the origin = **$\sqrt{(x^2 + y^2)}$**.

The formula is the Baudhāyana–Pythagoras theorem applied to the right triangle whose legs are the horizontal and vertical differences. **Example:** A(2, 3), B(5, 7): AB = √(9 + 16) = **5**. A(−3, −3), B(5, 3): AB = √(64 + 36) = **10**. The points (−5, −12) and (13, 0) are both at distance 13 from the origin.

1.4 Midpoint and Points of Trisection

FORMULA

Midpoint of AB = **$((x_1 + x_2)/2, (y_1 + y_2)/2)$**. **Point dividing AB in the ratio m : n** = $((m\,x_2 + n\,x_1)/(m + n), (m\,y_2 + n\,y_1)/(m + n))$.

- Midpoint of P(7, −4) and Q(−3, 2) = (2, −1).
- **Trisection points** of A(2, 5) and B(11, 14): the ratios are 1 : 2 and 2 : 1, giving (5, 8) and (8, 11).
- **Fourth vertex of a parallelogram:** the diagonals bisect each other, so the midpoints of the two diagonals are equal. For A(1, 2), B(4, 3), C(6, 6) the fourth vertex D satisfies A + C = B + D, so D = (3, 5).

1.5 Identifying Shapes by Coordinates

To show …	Method	Example
Isosceles triangle	two sides equal	A(−2, 0), B(4, 0), C(1, 4): AC = BC = 5
Equilateral triangle	all three sides equal	equal lengths
Right-angled triangle	$a^2 + b^2 = c^2$ for the sides	check the squares
Rectangle	opposite sides equal and diagonals equal	A(1, 1), B(5, 1), C(5, 4), D(1, 4): AB = DC = 4, AD = BC = 3, AC = BD = 5
Parallelogram	diagonals have the same midpoint	—
Collinear points	the sum of two distances equals the third	M(−3, −4), A(0, 0), G(6, 8): MA = 5, AG = 10, MG = 15
Equidistant point	set $PA^2 = PB^2$ and solve	On the y-axis equidistant from (6, 5) and (−4, 3): $36 + (y - 5)^2 = 16 + (y - 3)^2$, y = 9

IMPORTANT

Work with **squared distances** to avoid roots when comparing lengths. To prove collinearity, the sum of the two **shorter** distances must equal the **longest**.

Rectangle completion. Three vertices (2, 1), (8, 1), (8, 5): the fourth vertex is **(2, 5)**; the length is 6 and the breadth 4, so the area is 24 and the diagonal √52.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Distance between A(2, 3) and B(5, 7)?	$\sqrt{(9 + 16)} = 5$.
Midpoint of P(7, −4) and Q(−3, 2)?	(2, −1).
In which quadrant does (−2, 5) lie? What about (0, −4)?	Quadrant II; on the y-axis (no quadrant).
Reflection of (3, −2) in the x-axis?	(3, 2).
Is P(1, 0) equidistant from (2, 3) and (4, −1)?	$PA^2 = 10$ and $PB^2 = 10$; yes.
Show A(−2, 0), B(4, 0), C(1, 4) form an isosceles triangle.	AC = BC = 5, AB = 6.
Trisection points of (2, 5) and (11, 14)?	(5, 8) and (8, 11).
Show M(−3, −4), A(0, 0), G(6, 8) are collinear.	MA = 5, AG = 10, MG = 15 = 5 + 10.
Fourth vertex of a parallelogram with A(1, 2), B(4, 3), C(6, 6)?	D(3, 5).
Why are (3, 5) and (5, 3) different?	Order of the pair matters: the first is the x-coordinate.

★ KEY TAKEAWAYS

- **(x, y):** abscissa along the x-axis, ordinate along the y-axis; the axes divide the plane into four quadrants.
- Reflection in the **x-axis** changes the sign of y; in the **y-axis**, the sign of x.
- **Distance** = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$; **midpoint** = average of the coordinates.
- A **parallelogram's diagonals** share a midpoint; **collinear** points satisfy AB + BC = AC.
- Use **squared distances** to prove isosceles, right-angled, rectangle and equidistance results.

CHAPTER 02 · MATHS (PART 1)

Introduction to Linear Polynomials

Whenever a quantity changes by the same amount each step, a linear polynomial describes it. This chapter builds the nth-term rule of a linear pattern, names the parts of a polynomial, plots the graph y = ax + b as a straight line, interprets slope and intercept, finds the zero of a linear polynomial and models real situations.

Big Questions

- How do we write the rule of a pattern that grows by a constant amount?
- What do the slope and intercept of the line y = ax + b tell us?
- How do we form and solve linear equations from real situations?

2.1 Linear Patterns

A sequence with a **constant difference d** between terms follows a **linear rule**: first term a_1, nth term $\mathbf{a_n = a_1 + (n - 1)d}$. For 4, 7, 10, … the difference is 3, so the nth term is 3n + 1 and the 10th term is **31**. For 7, 11, 15, … the rule is 4n + 3, giving 4 × 15 + 3 = **63** as the 15th term.

Pattern	Difference	nth term
4, 7, 10, …	3	3n + 1
7, 11, 15, …	4	4n + 3
120, 111, 102, … (rally members, 9 leave each hour)	−9	129 − 9n (so 120 − 9t after t hours)
2, 5, 8, …	3	3n − 1

2.2 Polynomials and Linear Polynomials

DEFINITION

A **polynomial** in x is a sum of terms $a_n x^n$ with non-negative whole-number powers. The **degree** is the highest power of x. A **linear polynomial** has degree **1**: **ax + b** with a ≠ 0. The number a is the **coefficient** of x and b is the **constant term**.

Polynomial	Degree	Kind	Constant term
−2x + 6	1	linear	6
$9x^3 + 5x^2 - 8x - 10$	3	cubic	−10
$5x^2 - 3$	2	quadratic (**not** linear)	−3
7	0	constant	7
x + 1/x	—	**not a polynomial** (negative power)	—

The **value** of a polynomial at x is found by substitution: for p(x) = −2x + 6, p(1) = 4 and p(4) = −2. A student who calls $5x^2 - 3$ linear is wrong: the highest power is 2.

2.3 Graph of a Linear Polynomial

FORMULA

The graph of **y = mx + c** is a **straight line**. **m = slope** (the rise in y for a run of 1 in x); **c = y-intercept** (the line cuts the y-axis at (0, c)).

Feature	Meaning	Example y = 2x + 1
Slope m	m > 0 rising, m < 0 falling, m = 0 horizontal; larger |m| is steeper	m = 2: y rises 2 for every 1 step in x
Intercept c	value of y when x = 0	c = 1: passes through (0, 1)
Points on the line	satisfy the equation	(7, 15) lies on it since 2 × 7 + 1 = 15; (3, 8) does not as 2 × 3 + 1 = 7
Slope from two points	$(y_2 - y_1)/(x_2 - x_1)$	(0, 3) and (1, 5): slope 2

Lines y = x/2, y = x, y = 2x all pass through the origin; the larger the slope, the steeper the line. Two lines with the **same slope** are **parallel**.

2.4 Zero of a Linear Polynomial

DEFINITION

A **zero** of p(x) is a value of x for which p(x) = 0. For ax + b the zero is **x = −b/a**, the x-coordinate of the point where the line **cuts the x-axis**.

Zero of 3x − 12: x = 4. Zero of −2x + 6: x = 3. A linear polynomial has **exactly one** zero.

2.5 Modelling with Linear Polynomials

Situation	Model	Result
Ruby has 3 times as many ₹2 coins as ₹5 coins; total ₹88	5x + 2 × 3x = 11x = 88	8 five-rupee and 24 two-rupee coins
Temperature: ice 0 °C = 32 °F, water boils 100 °C = 212 °F	C = (5/9)(F − 32)	slope 5/9; at 50 °F, C = 10
Phone ₹10,000 loses ₹800 a year	V = 10,000 − 800t	after 3 years: ₹7,600
₹500 saved, ₹150 added each month	S = 500 + 150m	after 6 months: ₹1,400
Two numbers add to 64, one is 10 more than the other	x + (x + 10) = 64	27 and 37

REMEMBER

In a model the **slope** is the **rate of change** (₹150 per month, −₹800 per year) and the **intercept** is the **starting value** (₹500, ₹10,000).

Quick Revision: Likely Exam Questions

Likely question	Model answer
Write the nth term of 4, 7, 10, … and find the 10th.	3n + 1; 31.
What is a linear polynomial? Is $5x^2 - 3$ linear?	A polynomial of degree 1, ax + b; no, it has degree 2.
Value of −2x + 6 at x = 1 and x = 4?	4 and −2.
Equation of the line through (0, 3) with slope 2?	y = 2x + 3; cuts the y-axis at 3.
Show (7, 15) lies on y = 2x + 1 but (3, 8) does not.	2 × 7 + 1 = 15; 2 × 3 + 1 = 7 ≠ 8.
Zero of 3x − 12?	x = 4.
What does the slope represent?	The rate of change of y with x.
A phone worth ₹10,000 loses ₹800 yearly. Write its value after t years.	V = 10,000 − 800t.
Constant term of $9x^3 + 5x^2 - 8x - 10$?	−10.
What do the graphs of y = x/2, y = x and y = 2x show?	Straight lines through the origin that get steeper as the slope increases.

★ KEY TAKEAWAYS

- A **linear polynomial** has degree 1: **ax + b**; its graph is a **straight line**.
- **y = mx + c:** m is the **slope** (rate of change), c the **y-intercept** (starting value).
- The **zero** of ax + b is **−b/a** (where the line meets the x-axis).
- Constant difference d gives the nth term **$a_1 + (n - 1)d$**.
- Model situations by choosing the variable, writing the polynomial and **checking** the answer.

CHAPTER 03 · MATHS (PART 1)

The World of Numbers

Numbers have grown from counting numbers to a rich family. This chapter classifies the number families, studies closure, shows that rational numbers have terminating or repeating decimals and can be written as p/q, proves that numbers such as √2 and √5 are irrational and places rational and irrational numbers together on the real number line.

Big Questions

- How do natural numbers, integers and rationals extend one another and which operations are closed in each?
- How can a decimal be written as p/q, and when does a fraction terminate?
- What are irrational numbers and how do we know a number is irrational?

3.1 The Number Families

Set	Symbol	Members	Why it was needed
Natural numbers	N	1, 2, 3, …	counting
Whole numbers	W	0, 1, 2, 3, …	‘nothing’; Brahmagupta treated zero as a number (a − a = 0)
Integers	Z	…, −2, −1, 0, 1, 2, …	debts, subtraction always possible
Rational numbers	Q	p/q with p, q integers, q ≠ 0	division always possible (except by 0)
Real numbers	R	rationals and irrationals	lengths such as √2; every point of the number line

DEFINITION

A set is **closed** under an operation if the operation on any two members gives a member of the same set. N is closed under + and ×, **not** under − or ÷; Z is closed under +, −, ×, **not** ÷; Q is closed under +, −, × and ÷ (by a non-zero number).

- **Properties** of rationals: commutative and associative laws for + and ×; **distributive law** a(b + c) = ab + ac (check: 1/2 × (1/4 + 1/3) = 1/8 + 1/6 = 7/24); 0 is the additive identity, 1 the multiplicative identity; each rational has an additive inverse and each non-zero rational a reciprocal.
- Between any two rationals there are **infinitely many** rationals (the average of two rationals is a rational between them). Three between 1/3 and 1/2: 8/24 = 1/3 and 12/24 = 1/2, so 9/24, 10/24, 11/24.

3.2 Rational Numbers and Decimals

IMPORTANT

The decimal expansion of a rational number is either **terminating** or **non-terminating repeating**. A rational p/q in **lowest terms** has a **terminating** decimal **if and only if the prime factors of q are only 2 and/or 5**.

Fraction	Denominator factors	Decimal
11/20	$2^2 \times 5$	0.55 terminating
3/8	2^3	0.375 terminating, 3 places
7/16	2^4	0.4375 terminating
77/210 = 11/30	2 × 3 × 5 (has 3)	0.3666… repeating
1/7	7	0.142857 repeating, block of 6

If the denominator is $2^a \times 5^b$ the decimal terminates after **max(a, b)** places: 3/8 has 3 places, and a denominator $2^3 \times 5$ gives 3 places. The block 142857 repeats in the decimal of 1/7, 2/7, … (cyclic number).

▸ Writing a repeating decimal as p/q

Step 1
Let x = the repeating decimal, e.g. x = 0.666…

Step 2
Multiply by 10^n where n = length of the repeating block: 10x = 6.666…

Step 3
Subtract: 10x − x = 6, so 9x = 6 and **x = 2/3**.

More examples
0.181818… = 18/99 = **2/11** (multiply by 100); 0.1666…: 100x − 10x = 16.666… − 1.666… = 15, so x = 15/90 = **1/6**.

3.3 Irrational Numbers

DEFINITION

A number that **cannot** be written as p/q (q ≠ 0, p, q integers) is **irrational**. Its decimal expansion is **non-terminating and non-repeating**. Examples: √2, √3, √5, π, 0.1011011101111… (the pattern never repeats).

Prove √5 is irrational
Assume √5 = p/q in lowest terms (p, q have no common factor).

Square
$5q^2 = p^2$, so 5 divides p^2 and hence 5 divides p; write p = 5k.

Substitute
$5q^2 = 25k^2$, so $q^2 = 5k^2$ and 5 divides q.

Contradiction
p and q both divisible by 5 contradicts ‘lowest terms’. Hence √5 is irrational.

Number	Type	Reason
√81	rational	= 9
√12	irrational	= 2√3, √3 is irrational
0.333…	rational	= 1/3

Number	Type	Reason
2/√3	irrational	if it were rational r, then √3 = 2/r would be rational
√2 × √2	rational	= 2 (product of irrationals can be rational)
√2 + √3	irrational	—

NOTE

Sum, difference, product or quotient of a **non-zero rational and an irrational** is irrational (e.g. 3 + √2, 5√3). But the sum or product of **two irrationals** may be rational: √2 × √2 = 2, (2 + √3) + (2 − √3) = 4.

3.4 The Real Number Line

- Every real number corresponds to exactly **one point** on the number line and every point to one real number.
- To locate **3/4**: divide [0, 1] into 4 equal parts and take the third mark.
- To locate **√2**: draw a square of side 1 on the number line; its diagonal is √2; swing it with a compass to the line. Then a right triangle with legs √2 and 1 gives **√3**, and so on (the **square-root spiral**).
- Between any two real numbers there are infinitely many rationals **and** infinitely many irrationals.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Express 0.666… in the form p/q.	2/3.
Does 11/20 terminate?	Yes; $20 = 2^2 \times 5$, so 0.55.
Is the set of integers closed under division?	No; 1 ÷ 2 is not an integer.
Define an irrational number.	A number that cannot be written as p/q; non-terminating, non-repeating decimal.
Prove that √5 is irrational.	Assume √5 = p/q in lowest terms; p and q both divisible by 5, contradiction.
Find three rationals between 1/3 and 1/2.	9/24, 10/24, 11/24.
Classify √81, √12, 0.333….	Rational, irrational, rational.
Rational or irrational: √2 × √2?	Rational, equal to 2.
Which block repeats in 1/7?	142857.
How do we locate √2 on the number line?	Diagonal of the unit square marked with a compass from 0.

★ KEY TAKEAWAYS

- **N ⊂ W ⊂ Z ⊂ Q ⊂ R**; closure depends on the operation and the set.
- A rational has a **terminating** or **repeating** decimal; denominator with only 2 and 5 as prime factors means terminating.
- Convert a repeating decimal to p/q by **multiplying by 10^n and subtracting**.
- **Irrational:** non-terminating, non-repeating; √2, √5, π; proved by **contradiction**.
- Rationals and irrationals together fill the **real number line**.

CHAPTER 04 · MATHS (PART 1)

Exploring Algebraic Identities

An identity is an algebraic statement true for every value of the letters. This chapter proves the standard identities by area and by multiplication, extends them to three terms and to cubes, uses them to expand, factorise and calculate mentally, and meets Śhrīdharāchārya's old method for quick squaring.

Big Questions

- Which identities hold for squares, products and cubes and how are they proved?
- How do we use identities to factorise expressions?
- How do identities simplify arithmetic and algebraic fractions?

4.1 Identity or Equation?

DEFINITION

An **identity** is true for **all** values of its variables; an **equation** is true only for some. $(x + y)^2 = x^2 + 2xy + y^2$ is an identity; $x + 5 = 9$ is an equation (true only for $x = 4$). To **disprove** a claimed identity one counterexample is enough.

4.2 The Standard Identities

No.	Identity	Example
1	$\mathbf{(a + b)^2 = a^2 + 2ab + b^2}$	$(3x + 4)^2 = 9x^2 + 24x + 16$
2	$\mathbf{(a - b)^2 = a^2 - 2ab + b^2}$	$(2x - 3)^2 = 4x^2 - 12x + 9$
3	$\mathbf{(a + b)(a - b) = a^2 - b^2}$	$(7x + 2)(7x - 2) = 49x^2 - 4$
4	$\mathbf{(x + a)(x + b) = x^2 + (a + b)x + ab}$	$(x + 3)(x + 5) = x^2 + 8x + 15$
5	$\mathbf{(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca}$	$(3x + 2y + z)^2$
6	$\mathbf{(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3}$	$(x + 2)^3 = x^3 + 6x^2 + 12x + 8$
7	$\mathbf{(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3}$	$(2x - 1)^3 = 8x^3 - 12x^2 + 6x - 1$
8	$\mathbf{a^3 - b^3 = (a - b)(a^2 + ab + b^2)}$ and $\mathbf{a^3 + b^3 = (a + b)(a^2 - ab + b^2)}$	$27a^3 - 8b^3 = (3a - 2b)(9a^2 + 6ab + 4b^2)$
9	$\mathbf{x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)}$	if $x + y + z = 0$ then $x^3 + y^3 + z^3 = 3xyz$

▸ Why the identities are true

- **Area proof of $(a + b)^2$:** a square of side $a + b$ is cut into a^2, b^2 and two rectangles ab, so $(a + b)^2 = a^2 + b^2 + 2ab$.
- **$(a + b + c)^2$:** multiply $(a + b + c)(a + b + c)$; the squares appear once and each pair product appears **twice**, giving the 6 terms.

- **$(a + b)^3$:** multiply $(a + b)(a + b)^2$; or split a cube of edge $a + b$ into cubes a^3 and b^3 and **three** $a \times a \times b$ and **three** $a \times b \times b$ cuboids.
- Verify an identity with numbers: for $a = 2$, $b = 1$, $(a + b)^3 = 27$ and $8 + 12 + 6 + 1 = 27$ ✓.

> **REMEMBER**
>
> $(a + b)^2$ is **not** $a^2 + b^2$; the middle term **2ab** is essential. For $a = 2$, $b = 3$ the left side is 25 and $a^2 + b^2$ is 13.

4.3 Factorisation Using Identities

Perfect-square trinomials

$9x^2 + 24x + 16 = (3x)^2 + 2(3x)(4) + 4^2 = (3x + 4)^2$; $16y^2 - 24y + 9 = (4y - 3)^2$.

Difference of squares

$49x^2 - 25 = (7x + 5)(7x - 5)$.

Quadratic $x^2 + px + q$

find two numbers with **sum p** and **product q**: $x^2 - 7x + 12 = (x - 3)(x - 4)$; $r^2 - r - 42 = (r - 7)(r + 6)$.

Splitting the middle term $(ax^2 + bx + c)$

find numbers with sum b and product ac: $6x^2 + 7x + 2$: ac = 12, split $7x = 3x + 4x$, so $3x(2x + 1) + 2(2x + 1) = (2x + 1)(3x + 2)$.

Cubes

$8n^3 - 60n^2m + 150nm^2 - 125m^3 = (2n - 5m)^3$; $p^3 + 6p^2q + 12pq^2 + 8q^3 = (p + 2q)^3$ (the side of a cube).

Simplifying fractions: $(x^2 - 7x + 12)/(5x^2 + 5x - 100) = (x - 3)(x - 4)/[5(x - 4)(x + 5)] =$ **$(x - 3)/(5(x + 5))$**, for $x \neq 4$. Always state that the denominator is not zero.

4.4 Quick Arithmetic and Śhrīdharāchārya's Method

Calculation	Identity	Working	Answer
43^2	$(40 + 3)^2$	1600 + 240 + 9	1,849
99^2	$(100 - 1)^2$	10,000 − 200 + 1	9,801
104×96	$(100 + 4)(100 - 4)$	10,000 − 16	9,984
119^2	$(100 + 10 + 9)^2$	10,000 + 100 + 81 + 2,000 + 1,800 + 180	14,161
$105^2 + 95^2$	$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$	2(10,000 + 25)	20,050
103^3	$(100 + 3)^3$	10^6 + 90,000 + 2,700 + 27	1,092,727

> **FORMULA**
>
> **Śhrīdharāchārya (c. 750 CE):** $a^2 = (a + b)(a - b) + b^2$. Choose b so that $a + b$ or $a - b$ is a round number: **$55^2 = 60 \times 50 + 25 = 3{,}025$**, $65^2 = 70 \times 60 + 25 = 4{,}225$.

More results. The difference of the squares of two consecutive integers equals their sum: $(n + 1)^2 - n^2 = 2n + 1$. If $x + 1/x = 3$ then $x^2 + 1/x^2 = 9 - 2 =$ **7**. If $x + y + z = 10$ and $x^2 + y^2 + z^2 = 38$, then $xy + yz + zx = (100 - 38)/2 =$ **31**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Expand $(3x + 4)^2$.	$9x^2 + 24x + 16$.
Evaluate 99^2 using an identity.	$(100 - 1)^2 = 9{,}801$.
Factorise $49x^2 + 14xy + y^2$.	$(7x + y)^2$.
Factorise $x^2 - 5x + 6$.	$(x - 2)(x - 3)$.
State $(a + b)^3$ in expanded form.	$a^3 + 3a^2b + 3ab^2 + b^3$.
Is $(a + b)^2 = a^2 + b^2$ an identity?	No; the term $2ab$ is missing.
If $x + y + z = 0$, what is $x^3 + y^3 + z^3$?	$3xyz$.
Find 55^2 by Śhrīdharāchārya's method.	$60 \times 50 + 25 = 3{,}025$.
Factorise $27a^3 - 8b^3$.	$(3a - 2b)(9a^2 + 6ab + 4b^2)$.
Prove that $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$.	Expand and add: the $2ab$ terms cancel.

★ KEY TAKEAWAYS

- An **identity** holds for **all** values; one counterexample disproves it.
- **$(a \pm b)^2 = a^2 \pm 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$; $(x + a)(x + b) = x^2 + (a + b)x + ab$.**
- **$(a + b + c)^2$** has six terms: three squares and **$2ab + 2bc + 2ca$**.
- **Cubes:** $(a \pm b)^3$ and $a^3 \pm b^3$; if **$x + y + z = 0$**, $x^3 + y^3 + z^3 =$ **$3xyz$**.
- Use identities to **factorise** (sum–product method) and to calculate **mentally**.

CHAPTER 05 · MATHS (PART 1)

Circles

A circle is the set of all points at a fixed distance from a centre. This chapter studies chords and their distance from the centre, the relation between the angle an arc subtends at the centre and at the circumference, angles in a semicircle and in the same segment, and cyclic quadrilaterals and their angle properties.

Big Questions

- How are chords related to the centre of a circle?
- What is the relation between the angles an arc subtends at the centre and on the circle?
- What are the angle properties of a cyclic quadrilateral?

5.1 Basic Terms

Term	Meaning
Radius / diameter	centre to circle; chord through the centre (= 2r)
Chord	segment joining two points of the circle
Arc	part of the circle; **minor** arc < semicircle, **major** arc > semicircle
Segment	region between a chord and an arc
Sector	region between two radii and an arc
Cyclic quadrilateral	all four vertices lie on a circle

IMPORTANT

One and only one circle passes through three non-collinear points; through two points **infinitely many** circles pass; through three collinear points none does. The centre lies on the **perpendicular bisector** of every chord.

5.2 Chords and the Centre

Theorem	Idea of proof	Use
Equal chords subtend equal angles at the centre	congruent triangles by SSS (two radii and the chord)	chords AB = CD ⇒ ∠AOB = ∠COD
The perpendicular from the centre to a chord bisects it	RHS congruence of the two right triangles	half-chord = $\sqrt{(r^2 - d^2)}$
The line from the centre to the midpoint of a chord is perpendicular to it	converse by SSS	—
Equal chords are equidistant from the centre (and conversely)	right triangles with equal hypotenuse and half-chord	chords of equal length have equal d

Theorem	Idea of proof	Use
A longer chord is closer to the centre	$d = \sqrt{(r^2 - (c/2)^2)}$	diameter, the longest chord, has $d = 0$

FORMULA

For a chord of length **c** in a circle of radius **r**, the distance from the centre is $\mathbf{d = \sqrt{(r^2 - (c/2)^2)}}$, i.e. $(c/2)^2 + d^2 = r^2$.

Examples. A chord at distance 8 cm in a circle of radius 17 cm has half-length $\sqrt{(289 - 64)} = 15$, so its length is **30 cm**. Two chords of 10 cm in a circle of radius 13 cm are each at distance $\sqrt{(169 - 25)} =$ **12 cm**. Two parallel chords of 6 cm and 8 cm in a circle of radius 5 cm are at distances 4 cm and 3 cm; their separation is **7 cm** (same side: 1 cm).

5.3 Angles in a Circle

FORMULA

Angle at the centre theorem: the angle an arc subtends at the **centre** is **twice** the angle it subtends at any point on the **remaining part** of the circle. $\angle AOB = 2 \times \angle ACB$.

- An arc subtending 70° at the centre subtends **35°** at a point of the major arc. In ΔABC inscribed in a circle with centre O, if $\angle BAC = 50°$, then $\angle BOC =$ **100°**.
- **Angles in the same segment are equal:** all points on the same side of a chord see it at the same angle.
- **Angle in a semicircle = 90°:** the diameter subtends 180° at the centre, so 90° on the circle. Conversely, a right angle at C means AB is a diameter.
- If two angles on the **same side** of a segment are equal, the four points lie on a circle (**concyclic**).

5.4 Cyclic Quadrilaterals

DEFINITION

A **cyclic quadrilateral** has all four vertices on a circle. **Theorem:** the **opposite angles are supplementary** ($\angle A + \angle C = 180°$, $\angle B + \angle D = 180°$). **Converse:** if the opposite angles of a quadrilateral add to 180°, it is cyclic.

Proof idea. Join O to A and C. The arc ABC subtends $2\angle D$ at the centre and the arc ADC subtends $2\angle B$; the two angles at O add to 360°, so $2(\angle B + \angle D) = 360°$.

Result	Statement	Example
Exterior angle	the exterior angle equals the **interior opposite** angle	$\angle DCE = 70° \Rightarrow \angle A = 70°$
Opposite angles	sum 180°	$\angle P = (2x + 10)°$, $\angle R = (3x - 20)°$: $5x - 10 = 180$, $x = 38$
Cyclic trapezium	non-parallel sides are **equal** (isosceles trapezium)	$AB \parallel CD \Rightarrow AD = BC$
Cyclic parallelogram	must be a **rectangle**	opposite angles equal and add to $180° \Rightarrow 90°$

REMEMBER

To use the cyclic properties first **check that the four points lie on a circle**. A cyclic quadrilateral ABCD with sides AB = BC = 5 and CD = DA = 12 is a kite; being cyclic, its angles at A and C are 90° and the diagonal BD = √(25 + 144) = 13 is a diameter. Area = 2 × (½ × 5 × 12) = **60**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
A chord of a circle of radius 17 cm is 8 cm from the centre. Find its length.	2 × 15 = 30 cm.
An arc subtends 70° at the centre. Find the angle on the major arc.	35°.
PQRS is cyclic with ∠P = (2x + 10)° and ∠R = (3x − 20)°. Find x.	5x − 10 = 180, so x = 38.
State the angle sum of opposite angles of a cyclic quadrilateral.	180°.
Equal chords are at what distance from the centre?	Equal distances.
Through two points, how many circles can be drawn?	Infinitely many.
BC is produced to E in cyclic ABCD and ∠DCE = 70°. Find ∠A.	70°.
State the angle in a semicircle.	90°.
Prove that equal chords subtend equal angles at the centre.	Triangles AOB and COD are congruent by SSS.
ΔABC is inscribed with ∠BAC = 50°. Find ∠BOC.	100°.

★ KEY TAKEAWAYS

- **Perpendicular from the centre bisects the chord**; equal chords are equidistant from the centre.
- $d^2 + (c/2)^2 = r^2$ connects **radius, chord and distance**.
- **Angle at the centre = 2 × angle on the circle**; **angle in a semicircle = 90°**; angles in the same segment are equal.
- **Cyclic quadrilateral:** opposite angles add to **180°**; exterior angle = interior opposite angle.
- One circle passes through **three non-collinear** points.

CHAPTER 06 · MATHS (PART 1)

Measuring Space: Perimeter and Area

This chapter collects and extends the formulae for perimeter and area. It covers the circumference and area of a circle, the length of an arc and the area of a sector, areas of triangles (including Heron's formula), parallelograms and trapezia, and then applies them to rings and composite figures.

Big Questions

- How are the circumference and area of a circle related to π?
- How do we measure arcs, sectors, semicircles and quadrants?
- How do we find the area of a triangle from its three sides and of composite figures?

6.1 The Circle

FORMULA

Circumference $C = 2\pi r = \pi d$. Area $A = \pi r^2$. Here **$\pi = C/d \approx 22/7 \approx 3.14159\ldots$** is the constant ratio of circumference to diameter and is irrational. Use 22/7 when r is a multiple of 7.

Radius	Circumference (π = 22/7)	Area (π = 22/7)
7 cm	44 cm	154 cm^2
14 cm	88 cm	616 cm^2
21 cm	132 cm	1,386 cm^2

If the circumference is 44 cm, then 2πr = 44, so r = 44 × 7 ÷ 44 = **7 cm**. **Ring (annulus):** with outer radius R and inner radius r, area = **$\pi(R^2 - r^2)$**: for R = 14 and r = 7, area = 22/7 × (196 − 49) = **462 cm^2**.

6.2 Arc and Sector

FORMULA

For a sector of angle **θ** and radius r: **arc length = (θ/360) × 2πr**, **area of sector = (θ/360) × πr^2**, **perimeter of sector = 2r + arc length**.

Figure	Perimeter	Example
Semicircle (radius 7 cm)	πr + 2r	22 + 14 = 36 cm
Quadrant (radius 14 cm)	(πr/2) + 2r	22 + 28 = 50 cm
Sector, r = 14 cm, θ = 75°	arc + 2r	(75/360) × 88 = 18.33 + 28 = 46.33 cm
Sector, r = 21 cm, θ = 60°	arc + 2r	22 + 42 = 64 cm

Area of a sector with r = 21 cm and θ = 60°: (60/360) × (22/7) × 441 = **231 cm^2**. **Overlapping circles.** Two circles of radius r, each passing through the centre of the other, meet so that each common chord subtends 120° at its centre. The outline of the combined shape is made of two 240° arcs: (480/360) × 2πr = **8πr/3**.

6.3 Triangles

Triangle	Formula	Example
Any triangle	**½ × base × height**	base 9, height 4: 18
Equilateral (side a)	**$(\sqrt{3}/4)\,a^2$**	a = 8: $16\sqrt{3} \approx 27.7$
Isosceles (equal sides a, base b)	$\frac{1}{2} \times b \times \sqrt{(a^2 - b^2/4)}$	5, 5, 6: height 4, area 12
Right-angled	½ × product of the legs	legs 6 and 8: 24

FORMULA

Heron's formula: for sides a, b, c with **semi-perimeter s = (a + b + c)/2**, **Area = √[s(s − a)(s − b)(s − c)]**.

Example. Sides 13, 14, 15: s = 21; area = √(21 × 8 × 7 × 6) = √7056 = **84 cm^2**. The height on the side 14 is 2 × 84 ÷ 14 = **12 cm**. For an equilateral triangle s = 3a/2, so the area = $\sqrt{[(3a/2)(a/2)^3]} = (\sqrt{3}/4)a^2$.

6.4 Parallelograms and Trapezia

FORMULA

Parallelogram = base × height (perpendicular height, not the slant side). **Trapezium = ½ × (sum of parallel sides) × height. Rhombus = ½ × d_1 × d_2.**

- A parallelogram of area 96 cm^2 and base 12 cm has height **8 cm**.
- A trapezium with parallel sides 12 cm and 20 cm and height 7 cm has area ½ × 32 × 7 = **112 cm^2**.
- **Reason for the parallelogram formula:** cut off a right-angled triangle at one end and move it to the other end to get a rectangle of the same base and height.

6.5 Composite Figures

Step 1
Split the figure into rectangles, triangles, sectors or circles.

Step 2
Find each area (or perimeter) with the right formula and the same units.

Step 3
Add the pieces, or **subtract** the part cut out (e.g. a circular hole from a square plate).

Step 4
Give the answer in square units for area and in units for perimeter.

REMEMBER

For the **perimeter** of a composite figure add only the **outer boundary**; internal lines are not counted. Convert all measurements to a common unit first (1 m^2 = 10,000 cm^2).

Quick Revision: Likely Exam Questions

Likely question	Model answer
Circle perimeter 44 cm: find the radius (π = 22/7).	r = 7 cm.
Perimeter of a semicircular plate of radius 7 cm?	πr + 2r = 22 + 14 = 36 cm.
Area of a circle of radius 21 cm?	1,386 cm^2.
State Heron's formula.	√[s(s − a)(s − b)(s − c)] with s = (a + b + c)/2.
Area of the triangle with sides 13, 14, 15?	84 cm^2.
Area of an equilateral triangle of side a?	$(\sqrt{3}/4)a^2$.
Parallelogram area 96 cm^2, base 12 cm: height?	8 cm.
Area of a ring with radii 14 cm and 7 cm?	462 cm^2.
Trapezium with parallel sides 12 and 20, height 7: area?	112 cm^2.
What does π represent?	The ratio of circumference to diameter, same for every circle.

★ KEY TAKEAWAYS

- **$C = 2\pi r$, $A = \pi r^2$**; π is the circumference-to-diameter ratio.
- **Arc = $(\theta/360) \times 2\pi r$; sector area = $(\theta/360) \times \pi r^2$; sector perimeter = arc + 2r.**
- **Triangle ½bh; Heron $\sqrt{[s(s-a)(s-b)(s-c)]}$; equilateral $(\sqrt{3}/4)a^2$.**
- **Parallelogram bh; trapezium ½(a + b)h; rhombus ½d_1d_2.**
- Composite figures: **add or subtract** areas; count only the **outer boundary** for perimeter.

CHAPTER 07 · MATHS (PART 1)

The Mathematics of Maybe: Probability

Probability measures how likely an event is, as a number between 0 and 1. This chapter distinguishes experimental from theoretical probability, builds sample spaces for coins, dice, spinners and cards, uses complementary events, and explains why results settle down as the number of trials grows while individual outcomes remain unpredictable.

Big Questions

- How is the probability of an event measured by experiment and by reasoning?
- How do we list a sample space and count favourable outcomes?
- What does the law of large numbers say and why is the ‘gambler's fallacy’ wrong?

7.1 Basic Terms

Term	Meaning	Example (one die)
Experiment / trial	an action with uncertain result	rolling the die once
Outcome	one possible result	4
Sample space	the set of **all** outcomes	{1, 2, 3, 4, 5, 6}
Event	a collection of outcomes we are interested in	an even number = {2, 4, 6}
Equally likely	each outcome has the same chance	a fair die, a fair coin

7.2 Experimental Probability

FORMULA

Experimental (empirical) probability of an event E = (number of trials in which E occurs) ÷ (total number of trials).

A die rolled 50 times shows a 4 eight times: P(4) = 8/50 = **0.16**, compared with 1/6 ≈ 0.167. In 120 rolls with frequencies 18, 22, 19, 21, 24, 16 for 1 to 6, the experimental probability of a 5 is 24/120 = **0.2**. A shopkeeper who finds that 30 of 200 customers buy tea estimates P(tea) = 0.15 and expects about 150 tea buyers among 1,000 customers.

7.3 Theoretical Probability

FORMULA

For **equally likely** outcomes: **P(E) = (number of outcomes favourable to E) ÷ (total number of outcomes).** Always **0 ≤ P(E) ≤ 1**; P = 0 for an **impossible** event and P = 1 for a **certain** event.

Experiment	Sample space size	Event and probability
Toss one coin	2	P(head) = 1/2

Experiment	Sample space size	Event and probability
Toss two coins	4 (HH, HT, TH, TT)	P(exactly one head) = 2/4 = 1/2
Roll one die	6	P(prime) = 3/6 = 1/2 (2, 3, 5); P(even and prime) = 1/6
Roll two dice	36	P(same number) = 6/36 = 1/6; P(sum 7) = 6/36 = 1/6; P(sum is prime) = 15/36 = 5/12
Card from 52	52	P(king) = 4/52 = 1/13; P(heart) = 13/52 = 1/4; P(face card) = 12/52 = 3/13; P(ace) = 1/13
Spinner with 8 equal sectors	8	P(multiple of 3) = 2/8 = 1/4 (3, 6); P(prime) = 4/8 = 1/2 (2, 3, 5, 7)
A month of the year	12	P(31 days) = 7/12

NOTE

A standard pack has 4 suits (spades, hearts, diamonds, clubs; 13 cards each), 26 red and 26 black cards, 12 face cards (J, Q, K) and 4 aces.

7.4 Complementary Events

FORMULA

The **complement** of E (written ‘not E’) is the event that E does **not** occur. **P(E) + P(not E) = 1**, so **P(not E) = 1 − P(E).**

If P(E) = 5/12, then P(not E) = 7/12. If the probability of rain is 0.3 then the probability of no rain is 0.7. It is often easier to count the opposite: P(at least one head in two tosses) = 1 − P(no head) = 1 − 1/4 = **3/4**.

7.5 Law of Large Numbers and Misconceptions

- **Law of large numbers:** as the number of trials increases, the experimental probability tends to get **closer to the theoretical probability** (the fraction of heads in 10,000 tosses is nearer 1/2 than in 10 tosses).
- **Independence:** the coin has no memory. After six heads in a row the chance of tails on the next toss is still **1/2**.
- Believing that tails is ‘due’ is the **gambler's fallacy**.
- A probability of 1/6 does **not** mean that exactly one 4 will appear in each 6 rolls; it describes the long-run share.
- **Fair games** need equally likely outcomes (a fair die, a balanced coin).

IMPORTANT

Not every experiment has **equally likely** outcomes. The result of a cricket match has two outcomes (win or lose) but they are not equally likely, so P(win) is **not** 1/2. Use the experimental method when the outcomes are not equally likely.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Write the sample space for tossing two coins.	HH, HT, TH, TT.
Probability of exactly one head in two tosses?	2/4 = 1/2.
A die shows a 4 eight times in 50 rolls. Experimental probability of 4?	8/50 = 0.16.
A card is drawn from a pack of 52: P(king)?	4/52 = 1/13.
P(E) = 5/12. Find P(not E).	7/12.
Range of a probability?	0 to 1 inclusive.
Two dice are rolled: P(both show the same number)?	6/36 = 1/6.
Spinner with 8 equal sectors: P(multiple of 3)?	2/8 = 1/4.
A coin shows heads six times. Is tails certain next?	No; the probability is still 1/2.
State the law of large numbers.	With more trials the experimental probability approaches the theoretical probability.

★ KEY TAKEAWAYS

- **Experimental P = favourable trials ÷ total trials**; **theoretical P = favourable outcomes ÷ equally likely outcomes**.
- **0 ≤ P(E) ≤ 1**; impossible = 0, certain = 1.
- **P(E) + P(not E) = 1**: count the complement when it is easier.
- List the **sample space** carefully (two dice have 36 outcomes, a pack has 52 cards).
- **Law of large numbers:** long-run frequency settles near the theoretical value; past results do not change the next toss.

CHAPTER 08 · MATHS (PART 1)

Predicting What Comes Next: Sequences and Progressions

A sequence is an ordered list that follows a rule. This chapter describes sequences by explicit and recursive rules, studies the two most important progressions (arithmetic with constant difference and geometric with constant ratio), derives the nth term and the sum of an AP, and meets famous sequences such as the Virahānka numbers and the Sierpiński triangle.

Big Questions

- How do explicit and recursive rules describe a sequence?
- What are the nth term and sum formulae of an AP and the nth term of a GP?
- How do patterns such as the Virahānka–Fibonacci sequence and the Sierpiński triangle grow?

8.1 Sequences and Rules

DEFINITION

A **sequence** $t_1, t_2, t_3, \ldots$ is an ordered list of numbers; t_n is the **nth term**. A rule can be **explicit** (gives t_n directly from n) or **recursive** (gives t_n from the earlier terms and states the starting terms).

Sequence	Explicit rule	Recursive rule
1, 4, 7, 10, …	$t_n = 3n - 2$	$t_1 = 1,\ t_{n+1} = t_n + 3$
3, 6, 12, 24, …	$t_n = 3 \times 2^{n-1}$	$t_1 = 3,\ t_{n+1} = 2t_n$
1, 4, 9, 16, …	$t_n = n^2$	$t_{n+1} = t_n + 2n + 1$
1, 5, 13, 29, …	—	$u_1 = 1,\ u_n = 2u_{n-1} + 3$
1, 2, 3, 5, 8, …	—	$t_n = t_{n-1} + t_{n-2}$

Example. For $t_n = 3n - 4$ the first five terms are −1, 2, 5, 8, 11. For $t_1 = -5,\ t_{n+1} = t_n + 3$ we get −5, −2, 1, 4, 7, …; 52 is a term only if $-5 + 3(n - 1) = 52$, i.e. $n = 20$, so **yes** (the 20th term).

8.2 Arithmetic Progression (AP)

DEFINITION

An **AP** has a **constant difference d** between consecutive terms: a, a + d, a + 2d, … The first term is **a**.

FORMULA

nth term: $t_n = a + (n - 1)d$. Sum of the first n terms: $S_n = (n/2)[2a + (n - 1)d] = (n/2)$(first term + last term).

Problem	Working	Answer
Which term of 21, 18, 15, … is −81?	21 − 3(n − 1) = −81, n − 1 = 34	n = 35 (0 is the 8th term)
AP with 3rd term 12 and last (50th) term 106	a + 2d = 12 and a + 49d = 106, so d = 2, a = 8	29th term = 8 + 28 × 2 = 64
Marbles: 1, 2, 3, … 25 rows	S_{25} = 25/2 × (1 + 25)	325 marbles
Sum of the first n odd numbers	n/2 × (1 + (2n − 1))	n^2
Sum of 1 + 2 + … + 100 (Gauss)	100/2 × 101	5,050

NOTE

The points (n, t_n) of an AP lie on a **straight line**: for 2, 5, 8, … t_n = 3n − 1, a line of **slope d = 3**. This links progressions with linear polynomials (Chapter 2). An AP with **d > 0** increases, **d < 0** decreases, **d = 0** is constant.

8.3 Geometric Progression (GP)

DEFINITION

A **GP** has a **constant ratio r** between consecutive terms: a, ar, ar^2, … The nth term is $\mathbf{t_n = a\,r^{n-1}}$; recursively $t_{n+1} = r \times t_n$.

GP	a	r	Example term
3, 6, 12, …	3	2	$t_{10} = 3 \times 2^9 = 1{,}536$
2, 6, 18, 54, …	2	3	ratios 6/2 = 18/6 = 54/18 = 3
1, 1/2, 1/4, …	1	1/2	t_6 = 1/32 (decreasing)
5, −10, 20, …	5	−2	terms alternate in sign

REMEMBER

To test an AP, check **equal differences**; to test a GP, check **equal ratios**. An AP grows **linearly**; a GP grows (or shrinks) **exponentially**: 2^{10} = 1,024 but 2 × 10 = 20.

8.4 Famous Patterns

- **Virahānka–Fibonacci:** 1, 2, 3, 5, 8, 13, …, each term the sum of the two before. It was first described in India for counting rhythms of short and long syllables (Virahānka, Gopāla, Hemachandra), centuries before Fibonacci.
- **Tribonacci type:** $T_1 = 1$, $T_2 = 2$, $T_3 = 4$, $T_n = T_{n-1} + T_{n-2} + T_{n-3}$ gives 1, 2, 4, 7, 13, **24**.
- **Sierpiński triangle:** start with a black triangle (Stage 0) and remove the central triangle at each stage. The number of black triangles is $\mathbf{3^n}$ (Stage 4: 81, Stage 5: 243) and the black area is $\mathbf{(3/4)^n}$ of the original (Stage 3: 27/64): a GP with ratio 3/4.
- **Square numbers** 1, 4, 9, 16, …: not an AP (differences 3, 5, 7, … form an AP) and not a GP; the second differences are constant (2).

Quick Revision: Likely Exam Questions

Likely question	Model answer
First five terms of $t_n = 3n - 4$?	−1, 2, 5, 8, 11.
10th and 15th terms of $t_n = 5n - 3$?	47 and 72.
Which term of 21, 18, 15, … is −81?	35th term.
Show that 2, 6, 18, 54, … is a GP and write its nth term.	Ratio 3; $t_n = 2 \times 3^{n-1}$.
Sum of the first 25 natural numbers?	25 × 26/2 = 325.
Write the explicit and recursive rule of 1, 4, 7, 10, ….	$t_n = 3n - 2$; $t_1 = 1$, $t_{n+1} = t_n + 3$.
10th term of the GP 3, 6, 12, …?	1,536.
Number of black triangles at Stages 4 and 5 of the Sierpiński triangle?	81 and 243.
First five terms of $u_1 = 1$, $u_n = 2u_{n-1} + 3$; is 133 a term?	1, 5, 13, 29, 61, 125, 253; no, 133 is not a term.
What type of growth do an AP and a GP show?	AP: linear (constant difference); GP: exponential (constant ratio).

★ KEY TAKEAWAYS

- A sequence can have an **explicit** rule (t_n in terms of n) or a **recursive** rule (t_n from earlier terms).
- **AP:** $t_n = a + (n - 1)d$, $S_n = (n/2)[2a + (n - 1)d]$; its graph points lie on a **line**.
- **GP:** $t_n = a\, r^{n-1}$ with constant ratio r; grows or shrinks **exponentially**.
- **Virahānka–Fibonacci** numbers: each term is the sum of the previous two.
- **Sierpiński triangle:** 3^n black triangles, black area $(3/4)^n$.